

3-1-1a

1. If f and g are twice differentiable functions such that $g(x) = e^{f(x)}$ and $g''(x) = h(x)e^{f(x)}$, then $h(x) =$
- (A) $f'(x) + f''(x)$
- (B) $f'(x) + (f''(x))^2$
- (C) $(f'(x) + f''(x))^2$
- (D) $(f'(x))^2 + f''(x)$ ✓
- (E) $2f'(x) + f''(x)$
2. If $y = \left(\frac{x}{x+1}\right)^5$, then $\frac{dy}{dx} =$
- (A) $5(1+x)^4$
- (B) $\frac{x^4}{(x+1)^4}$
- (C) $\frac{5x^4}{(x+1)^4}$
- (D) $\frac{5x^4}{(x+1)^6}$ ✓
- (E) $\frac{5x^4(2x+1)}{(x+1)^6}$
3. If $f(x) = (x^2 - 3)^4$, then $f'(1) =$
- (A) -64 ✓
- (B) -32
- (C) -16
- (D) 32

Answer A

This option is correct. Both the power rule and the chain rule must be used to compute the derivative, which is then evaluated at $x = 1$.

$$f'(x) = 4(x^2 - 3)^3 (2x)$$

$$f'(1) = 4(1 - 3)^3 (2) = 4(-2)^3 (2) = -64$$

3-1-1a

4. If $y = (x^3 + 1)^2$, then $\frac{dy}{dx} =$

(A) $(3x^2)^2$

(B) $2(x^3 + 1)$

(C) $2(3x^2 + 1)$

(D) $3x^2(x^3 + 1)$

(E) $6x^2(x^3 + 1)$



3-1-1a

5.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

A light is directly above the western end of the track. Ben rides so that at time t , the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t=40$?

Part D

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

1 point is earned for units in (a) or (b)

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

1 point is earned for units in (a) or (b)

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

3-1-1a

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the family of functions $f(x) = \frac{1}{x^2 - 2x + k}$, where k is a constant.

a) Find the value of k , for $k > 0$, such that the slope of the line tangent to the graph of f at $x = 0$ equals 6.

b) For $k = -8$, find the value of $\int_0^1 f(x) \, dx$.

c) For $k = 1$, find the value of $\int_0^2 f(x) \, dx$ or show that it diverges.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a denominator of $(x^2 - 2x + k)^2$ or a factor of $(x^2 - 2x + k)^{-2}$.

The second point is earned for $f'(x) = \frac{-(2x-2)}{(x^2-2x+k)^2}$. A response that calculates the derivative correctly but then simplifies incorrectly will earn the second point, but is not eligible for the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ denominator of $f'(x)$
- ☐ $f'(x)$
- ☐ answer

Solution:

3-1-1a

$$f'(x) = \frac{-(2x-2)}{(x^2-2x+k)^2}$$

$$f'(0) = \frac{2}{k^2} = 6 \Rightarrow k^2 = \frac{1}{3} \Rightarrow k = \frac{1}{\sqrt{3}}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for obtaining $A = \frac{1}{6}$ and $B = -\frac{1}{6}$ with denominators $x - 4$ and $x + 2$, respectively. Other nonzero numerators with these denominators will not earn the first point, but the response is eligible for the second and third points.

A factorization of $(x + 4)(x - 2)$ resulting in nonzero numerators is eligible to earn the second point, but not the third point.

Both correct antiderivatives, including use of absolute value, earn the second point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ partial fraction decomposition
- ☐ antiderivatives
- ☐ answer

Solution:

$$\frac{1}{x^2-2x-8} = \frac{1}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}$$

$$\Rightarrow 1 = A(x+2) + B(x-4)$$

$$\Rightarrow A = \frac{1}{6}, B = -\frac{1}{6}$$

$$\int_0^1 f(x) dx = \int_0^1 \left(\frac{\frac{1}{6}}{x-4} - \frac{\frac{1}{6}}{x+2} \right) dx$$

$$= \left[\frac{1}{6} \ln |x-4| - \frac{1}{6} \ln |x+2| \right]_{x=0}^{x=1}$$

$$= \left(\frac{1}{6} \ln 3 - \frac{1}{6} \ln 3 \right) - \left(\frac{1}{6} \ln 4 - \frac{1}{6} \ln 2 \right) = -\frac{1}{6} \ln 2$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1a

The first point is earned for the definite integral $\int_0^2 \frac{1}{x^2 - 2x + 1} dx$ split into two integrals at $x = 1$, or work that shows that $f(x)$ has a vertical asymptote or infinite discontinuity at $x = 1$.

If the first point is not earned, the response is only eligible for the second point.

The second point requires a correct antiderivative with or without reference to a substitution. Correct limits of integration are not required for the second point.

The third point requires evidence of an improper integral, including use of appropriate limit notation. The third point is earned with a correctly evaluated one-sided limit along with the conclusion of divergence. Only one of the two integrals need to be referenced. Note that any limit that “ $= \frac{1}{0}$ ” makes the response not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ improper integral
- ☐ antiderivative
- ☐ answer with reason

Solution:

$$\begin{aligned}
 \int_0^2 \frac{1}{x^2 - 2x + 1} dx &= \int_0^2 \frac{1}{(x-1)^2} dx = \int_0^1 \frac{1}{(x-1)^2} dx + \int_1^2 \frac{1}{(x-1)^2} dx \\
 &= \lim_{b \rightarrow 1^-} \int_0^b \frac{1}{(x-1)^2} dx + \lim_{b \rightarrow 1^+} \int_b^2 \frac{1}{(x-1)^2} dx \\
 &= \lim_{b \rightarrow 1^-} \left(-\frac{1}{x-1} \Big|_{x=0}^{x=b} \right) + \lim_{b \rightarrow 1^+} \left(-\frac{1}{x-1} \Big|_{x=b}^{x=2} \right) \\
 &= \lim_{b \rightarrow 1^-} \left(-\frac{1}{b-1} - 1 \right) + \lim_{b \rightarrow 1^+} \left(-1 + \frac{1}{b-1} \right)
 \end{aligned}$$

Because $\lim_{b \rightarrow 1^-} \left(-\frac{1}{b-1} \right)$ does not exist, the integral diverges.

3-1-1a

7.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	6
$f(x)$	-5	4	1	7
$f'(x)$	-1	2	-2	4

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(x^2 - x)$.

- (a) What is the value of $g'(3)$?
- (b) It is known that $g''(0) = -1$. What is the value of $f''(0)$?
- (c) Is there a value c , for $0 < c < 3$, such that $g(c) = 5$? Justify your answer.
- (d) Let h be the function with derivative given by $h'(x) = 4e^{\cos x}$. At what value of x in the interval $-3 \leq x \leq 0$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $-3 \leq x \leq 0$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

3-1-1a



0	1	2
---	---	---

The student response accurately includes one of the criteria below.

☐ $g'(x)$

☐ $g'(3)$

Solution:

$$g'(x) = f'(x^2 - x)(2x - 1)$$

$$g'(3) = f'(6)(5) = 4(5) = 20$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

☐ $g''(x)$ (2 points for full credit)

☐ answer

Solution:

$$g''(x) = f''(x^2 - x)(2x - 1)^2 + f'(x^2 - x)(2)$$

$$g''(0) = f''(0)(-1)^2 + f'(0)(2)$$

$$-1 = f''(0) + 2(2) \Rightarrow f''(0) = -5$$

Part C

3-1-1a

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous, with reason
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = 4 \text{ and}$$

$$g(3) = f(6) = 7.$$

$$\Rightarrow g(0) < 5 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 5$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

3-1-1a

The average rate of change of f over the interval $-3 \leq x \leq 0$

$$\frac{f(0)-f(-3)}{0-(-3)} = \frac{4-(-5)}{3} = 3.$$

$$h'(x) = 4e^{\cos x} = 3 \text{ for } -3 \leq x \leq 0. \Rightarrow x = -1.862602$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $-3 \leq x \leq 0$ at $x = -1.863$ (or -1.862).

3-1-1a

8. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	1	2
$g(x)$	6	4	2	-1
$g'(x)$	-1	-7	-2	-3

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(1) = 0$ and derivative given by $f'(x) = x \sin(x^2)$.

- Find $f''(e^{1.5})$. Express your answer as a decimal approximation.
- Let h be the function defined by $h(x) = f(g(3x))$. Find $h'(0)$. Express your answer as a decimal approximation.
- Write an equation for the line tangent to the graph of g^{-1} , the inverse function of g , at $x = -1$.
- The point $(1, 4)$ lies on the curve in the xy -plane given by the equation $f(x) + (g(x))^2 = xy$. What is the value of $\frac{dy}{dx}$ at the point $(1, 4)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0

1

3-1-1a

The student response accurately includes a correct decimal approximation of $f''(e^{1.5})$

Solution:

$$f''(e^{1.5}) = 14.144$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = f'(g(3x)) \cdot g'(3x) \cdot 3$$

$$\begin{aligned} h'(0) &= f'(g(0)) \cdot g'(0) \cdot 3 = f'(4) \cdot (-7) \cdot 3 \\ &= 4 \sin 16 \cdot (-21) = -84 \sin 16 \\ &= 24.184 \text{ (or } 24.183) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

3-1-1a

- ☐ slope
- ☐ equation for tangent line

Solution:

Because $g(2) = -1$, $g^{-1}(-1) = 2$.

$$(g^{-1})'(-1) = \frac{1}{g'(g^{-1}(-1))} = \frac{1}{g'(2)} = \frac{1}{-3}$$

An equation for the tangent line is $y = 2 - \frac{1}{3}(x + 1)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error.

Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}(f(x) + (g(x))^2) = \frac{d}{dx}(xy)$$

$$\Rightarrow f'(x) + 2g(x) \cdot g'(x) = y + x \frac{dy}{dx}$$

$$\Rightarrow f'(1) + 2g(1) \cdot g'(1) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \sin 1 + 2 \cdot 2 \cdot (-2) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \sin 1 - 12 = -11.159 \text{ (or } -11.158)$$

3-1-1a

9.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	4
$f(x)$	5	-1	2	7
$f'(x)$	-2	4	0	1

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(2x - x^2)$.

- What is the value of $g'(-1)$?
- It is known that $g''(0) = 0$. What is the value of $f''(0)$?
- Is there a value c , for $0 < c < 3$, such that $g(c) = 2$? Justify your answer.
- Let h be the function with first derivative given by $h'(x) = 4xe^x$. At what value of x in the interval $0 \leq x \leq 4$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $0 \leq x \leq 4$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

3-1-1a



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(-1)$

Solution:

$$g'(x) = f'(2x - x^2)(2 - 2x)$$

$$g'(-1) = f'(-3)(4) = (-2)(4) = -8$$

Part B

Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(2x - x^2)(2 - 2x)^2 + f'(2x - x^2)(-2)$$

$$g''(0) = f''(0)(2)^2 + f'(0)(-2)$$

$$0 = 4f''(0) + 4(-2) \Rightarrow f''(0) = 2$$

Part C

3-1-1a

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ g is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = -1 \text{ and } g(3) = f(-3) = 5.$$

$$\Rightarrow g(0) < 2 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 2$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

3-1-1a

The average rate of change of f over the interval $0 \leq x \leq 4$ is $\frac{f(4)-f(0)}{4-0} = \frac{7-(-1)}{4} = 2$.

$$h'(x) = 4xe^x = 2 \text{ for } 0 \leq x \leq 4. \Rightarrow x = 0.351734$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $0 \leq x \leq 4$ at $x = 0.352$ (or 0.351).

3-1-1a

10. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	2	3
$g(x)$	4	1	0	-2
$g'(x)$	-2	-6	-3	-4

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(3) = 2$ and derivative given by $f'(x) = x \cos(e^x)$.

- Find $f''(\sin 2)$. Express your answer as a decimal approximation.
- Let h be the function defined by $h(x) = g(f(2x))$. Find $h'(1.5)$. Express your answer as a decimal approximation.
- Write an equation for the line tangent to the graph of g^{-1} , the inverse function of g , at $x = -2$.
- The point $(3, 1)$ lies on the curve in the xy -plane given by the equation $(f(x))^2 + g(x) = xy - 1$. What is the value of $\frac{dy}{dx}$ at the point $(3, 1)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0

1

3-1-1a

The student response accurately includes a correct decimal approximation of $f''(\sin 2)$

Solution:

$$f''(\sin 2) = -2.173 \text{ (or } -2.172)$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = g'(f(2x)) \cdot f'(2x) \cdot 2$$

$$\begin{aligned} h'(1.5) &= g'(f(3)) \cdot f'(3) \cdot 2 = g'(2) \cdot (3 \cos(e^3)) \cdot 2 \\ &= -3(3 \cos(e^3)) \cdot 2 \\ &= -18 \cos(e^3) = -5.915 \text{ (or } -5.914) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

3-1-1a**Solution:**

Because $g(3) = -2$, $g^{-1}(-2) = 3$.

$$(g^{-1})'(-2) = \frac{1}{g'(g^{-1}(-2))} = \frac{1}{g'(3)} = -\frac{1}{4}$$

An equation for the tangent line is $y = 3 - \frac{1}{4}(x + 2)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error. Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}((f(x))^2 + g(x)) = \frac{d}{dx}(xy - 1)$$

$$\Rightarrow 2f(x) \cdot f'(x) + g'(x) = y + x \frac{dy}{dx}$$

$$\Rightarrow 2f(3) \cdot f'(3) + g'(3) = 1 + 3 \frac{dy}{dx}$$

$$\Rightarrow 2(2) \cdot (3 \cos(e^3)) + (-4) = 1 + 3 \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -0.352$$

3-1-1a

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.1	2.2	2.3	2.4
$g'(x)$	16	25	36	49	64

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 5$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 2$ to approximate $g(2.2)$.
- Find the average rate of change of g' over the interval $2 \leq x \leq 2.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = (g(x))^3$. Find $h'(2)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_2^{2.4} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $2 \leq x \leq 2.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_2^{2.4} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.2)$. Show the computations that lead to your answer.
- It is known that $g''(2) = 80$ and $g'''(2) = 200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $5 + 16 \cdot 0.2$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.

3-1-1a



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(2.2) \approx g(2) + g'(2)(2.2 - 2) = 5 + 16 \cdot 0.2 = 8.2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{64-16}{0.4}$ or $\frac{48}{2.4-2}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(2.4) - g'(2)}{2.4 - 2} = \frac{64 - 16}{0.4} = 120$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $3 \cdot 5^2 \cdot 16$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = 3(g(x))^2 \cdot g'(x)$.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

3-1-1a

The answer point is earned only by the correct answer $3 \cdot 5^2 \cdot 16$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 3 \cdot (g(x))^2 \cdot g'(x)$$

$$h'(2) = 3 \cdot (g(2))^2 \cdot g'(2) = 3 \cdot 5^2 \cdot 16 = 1200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $25 \cdot 0.2 + 49 \cdot 0.2$ or $(25 + 49) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned} \int_2^{2.4} g'(x) \, dx &\approx g'(2.1)(2.2 - 2) + g'(2.3)(2.4 - 2.2) \\ &= 25 \cdot 0.2 + 49 \cdot 0.2 = 14.8 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1a

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_2^{2.4} g'(x) dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

“Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

“Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

“Since $g'(x)$ is increasing and $g'''(x) > 0$, is greater than 0, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

“ $g'(x)$ is increasing. Since $g'''(x) > 0$, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ ” is not part of the necessary conditions.

“Since $g'''(x) > 0$, $g'(x)$ is concave up. Since it is also increasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

“The approximation is an underestimate because $g'(x)$ is increasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **one** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) > 0$ for $2 \leq x \leq 2.4$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_2^{2.4} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $5 + 16 \cdot 0.1 = 6.6$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 9.1 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $5 + 16 \cdot 0.1 + 25 \cdot 0.1$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
2	5	16	0.1	1.6	6.6
2.1	6.6	25	0.1	2.5	9.1

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.1) \approx g(2) + g'(2)(2.1 - 2) = 5 + 16 \cdot 0.1 = 6.6$$

$$g(2.2) \approx g(2.1) + g'(2.1)(2.2 - 2.1) \approx 6.6 + 25 \cdot 0.1 = 9.1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 2$.

Because $g''(2) = 80$ and $g'''(2) = 200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(2)$ and $g'''(2)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(2) = 5$ given in the initial statement of the problem and $g'(2) = 16$ from the table for the coefficients involving those values. For example:

The expression $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns only one point.

The expression $5 + 16(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names (t , x , y , ...) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 2$ will earn no points.



0	1	2
---	---	---

3-1-1a

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ = 5 + 16(x-2) + \frac{80}{2}(x-2)^2 + \frac{200}{6}(x-2)^3 \\ = 5 + 16(x-2) + 40(x-2)^2 + \frac{100}{3}(x-2)^3 \end{aligned}$$

3-1-1a

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.1	2.2	2.3	2.4
$g'(x)$	20	39	56	71	84

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 9$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 2$ to approximate $g(2.4)$.
- Find the average rate of change of g' over the interval $2 \leq x \leq 2.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 6 + (g(x))^2$. Find $h'(2)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_2^{2.4} g'(x) \, dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) < 0$ for $2 \leq x \leq 2.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_2^{2.4} g'(x) \, dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.4)$. Show the computations that lead to your answer.
- It is known that $g''(2) = 200$ and $g'''(2) = -200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $9 + 20 \cdot 0.4$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.

3-1-1a



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(2.4) \approx g(2) + g'(2)(2.4 - 2) = 9 + 20 \cdot 0.4 = 17$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{84-20}{0.4}$ or $\frac{64}{2.4-2}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(2.4) - g'(2)}{2.4 - 2} = \frac{84 - 20}{0.4} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $2 \cdot 9 \cdot 20$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = 2 \cdot g(x) \cdot g'(x)$.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

3-1-1a

The answer point is earned only by the correct answer $2 \cdot 9 \cdot 20$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Chain rule

☐ Answer

Solution:

$$h'(x) = 2 \cdot g(x) \cdot g'(x)$$

$$h'(2) = 2 \cdot g(2) \cdot g'(2) = 2 \cdot 9 \cdot 20 = 360$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $39 \cdot 0.2 + 71 \cdot 0.2$ or $(39 + 71) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\begin{aligned} \int_2^{2.4} g'(x) dx &\approx g'(2.1)(2.2 - 2) + g'(2.3)(2.4 - 2.2) \\ &= 39 \cdot 0.2 + 71 \cdot 0.2 = 22 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1a

General:

The first point is for concluding that $g'(x)$ is concave down given that $g'''(x)$ is negative.

The answer point is earned for using that $g'(x)$ is concave down on the given interval to conclude that the midpoint Riemann sum approximation of $\int_2^{2.4} g'(x) dx$ computed in part (d) must necessarily be an overestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an underestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

· “Since $g'''(x) < 0$, then $g''(x)$ is decreasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

· “Since $g'''(x) < 0$, the tangent line to the graph of $g'(x)$ lies above $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

· “Since $g'(x)$ is increasing and $g'''(x) < 0$, $g'(x)$ is concave down.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

· “ $g'(x)$ is increasing. Since $g'''(x) < 0$, then $g'(x)$ is concave down.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.

· “Since $g'''(x) < 0$, $g'(x)$ is concave down. Since it is also increasing, the approximation is an overestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

· “The approximation is an overestimate because $g'(x)$ is increasing and concave down.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) < 0$ for $2 \leq x \leq 2.4$, g' is concave down on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_2^{2.4} g'(x) dx$ is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $9 + 20 \cdot 0.2 = 13$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 24.2 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $9 + 20 \cdot 0.2 + 56 \cdot 0.2$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
2	9	20	0.2	4	13
2.2	13	56	0.2	11.2	24.2

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.2) \approx g(2) + g'(2)(2.2 - 2) = 9 + 20 \cdot 0.2 = 13$$

$$g(2.4) \approx g(2.2) + g'(2.2)(2.4 - 2.2) \approx 13 + 56 \cdot 0.2 = 24.2$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 2$.

Because $g''(2) = 200$ and $g'''(2) = -200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(2)$ and $g'''(2)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(2) = 9$ given in the initial statement of the problem and $g'(2) = 20$ from the table for the coefficients involving those values. For example:

· The expression $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns only one point.

· The expression $9 + 20(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names ($t, x, y, \dots$) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 2$ will earn no points.



0	1	2
---	---	---

3-1-1a

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ = 9 + 20(x-2) + \frac{200}{2}(x-2)^2 + \frac{-200}{6}(x-2)^3 \\ = 9 + 20(x-2) + 100(x-2)^2 - \frac{100}{3}(x-2)^3 \end{aligned}$$

3-1-1a

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.2	3.4	3.6	3.8
$g'(x)$	10	20	34	52	74

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 7$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 3$ to approximate $g(3.4)$.
- Find the average rate of change of g' over the interval $3 \leq x \leq 3.8$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 4(g(x))^2$. Find $h'(3)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_3^{3.8} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $3 \leq x \leq 3.8$. Is the approximation from part (d) an overestimate or an underestimate for $\int_3^{3.8} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.4)$. Show the computations that lead to your answer.
- It is known that $g''(3) = 40$ and $g'''(3) = 100$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $7 + 10 \cdot 0.4$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.

3-1-1a



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3.4) \approx g(3) + g'(3)(3.4 - 3) = 7 + 10 \cdot 0.4 = 11$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{74-10}{0.8}$ or $\frac{64}{3.8-3}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(3.8) - g'(3)}{3.8 - 3} = \frac{74 - 10}{0.8} = 80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $8 \cdot 7 \cdot 10$.

The first point is for the correct application of the chain rule to produce the derivative $h'(x) = c \cdot 2 \cdot g(x) \cdot g'(x)$, where c is a non-zero constant. For example:

$\cdot 2 \cdot g(x) \cdot g'(x)$ earns the first point

3-1-1a

$\cdot 8 \cdot g(x) \cdot g'(x)$ earns the first point.

$\cdot 3 \cdot g(x) \cdot g'(x)$ does not earn the first point since it is not of the correct form.

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $8 \cdot 7 \cdot 10$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = 8 \cdot g(x) \cdot g'(x)$$

$$h'(3) = 8 \cdot g(3) \cdot g'(3) = 8 \cdot 7 \cdot 10 = 560$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $20 \cdot 0.4 + 52 \cdot 0.4$ or $(20 + 52) \cdot 0.4$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

3-1-1a

$$\int_3^{3.8} g'(x) dx \approx g'(3.2)(3.4 - 3) + g'(3.6)(3.8 - 3.4)$$

$$= 20 \cdot 0.4 + 52 \cdot 0.4 = 28.8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_3^{3.8} g'(x) dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

“Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

“Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

“Since $g'(x)$ is increasing and $g'''(x) > 0$, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.

“ $g'(x)$ is increasing. Since $g'''(x) > 0$, is greater than 0, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.

3-1-1a

“Since $g'''(x) > 0$, is greater than 0, $g'(x)$ is concave up. Since it is also increasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

“The approximation is an underestimate because $g'(x)$ is increasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) > 0$ for $3 \leq x \leq 3.8$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_3^{3.8} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $7 + 10 \cdot 0.2 = 9$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 13 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $7 + 10 \cdot 0.2 + 20 \cdot 0.2$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

3-1-1a

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
3	7	10	0.2	2	9
3.2	9	20	0.2	4	13

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.2) \approx g(3) + g'(3)(3.2 - 3) = 7 + 10 \cdot 0.2 = 9$$

$$g(3.4) \approx g(3.2) + g'(3.2)(3.4 - 3.2) \approx 9 + 20 \cdot 0.2 = 13$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 3$.

Because $g''(3) = 40$ and $g'''(3) = 100$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(3)$ and $g'''(3)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(3) = 7$ given in the initial statement of the problem and $g'(3) = 10$ from the table for the coefficients involving those values. For example:

The expression $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns only one point.

The expression $7 + 10(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names ($t, x, y, \dots$) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

3-1-1a

Taylor polynomials that are centered at a point other than $x = 3$ will earn no points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} &g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ &= 7 + 10(x-3) + \frac{40}{2}(x-3)^2 + \frac{100}{6}(x-3)^3 \\ &= 7 + 10(x-3) + 20(x-3)^2 + \frac{50}{3}(x-3)^3 \end{aligned}$$

3-1-1a

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	4	4.2	4.4	4.6	4.8
$g'(x)$	70	48	30	16	6

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(4) = 5$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 4$ to approximate $g(4.8)$.
- Find the average rate of change of g' over the interval $4 \leq x \leq 4.8$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = 1 + (g(x))^2$. Find $h'(4)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_4^{4.8} g'(x) \, dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) > 0$ for $4 \leq x \leq 4.8$. Is the approximation from part (d) an overestimate or an underestimate for $\int_4^{4.8} g'(x) \, dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 4$ with two steps of equal size, to approximate $g(4.8)$. Show the computations that lead to your answer.
- It is known that $g''(4) = -120$ and $g'''(4) = 100$. Write the third-degree Taylor polynomial for g about $x = 4$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $5 + 70 \cdot 0.8$.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.

3-1-1a



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(4.8) \approx g(4) + g'(4)(4.8 - 4) = 5 + 70 \cdot 0.8 = 61$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{6-70}{0.8}$ or $\frac{-64}{4.8-4}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(4.8) - g'(4)}{4.8 - 4} = \frac{6 - 70}{0.8} = -80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $2 \cdot 5 \cdot 70$.

The first point is for the correct application of the chain rule on the term $(g(x))^2$ to produce the derivative $2 \cdot g(x) \cdot g'(x)$. If there are any errors in differentiating the constant term, the student does not earn the second point.

3-1-1a

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $2 \cdot 5 \cdot 70$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = 2 \cdot g(x) \cdot g'(x)$$

$$h'(4) = 2 \cdot g(4) \cdot g'(4) = 2 \cdot 5 \cdot 70 = 700$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $48 \cdot 0.4 + 16 \cdot 0.4$ or $(48 + 16) \cdot 0.4$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} \int_4^{4.8} g'(x) dx &\approx g'(4.2)(4.4 - 4) + g'(4.6)(4.8 - 4.4) \\ &= 48 \cdot 0.4 + 16 \cdot 0.4 = 25.6 \end{aligned}$$

Part E

3-1-1a

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave up given that $g'''(x)$ is positive.

The answer point is earned for using that $g'(x)$ is concave up on the given interval to conclude that the midpoint Riemann sum approximation of $\int_4^{4.8} g'(x) dx$ computed in part (d) must necessarily be an underestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an overestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

· “Since $g'''(x) > 0$, then $g''(x)$ is increasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.

· “Since $g'''(x) > 0$, the tangent line to the graph of $g'(x)$ lies below $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is decreasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

· “Since $g'(x)$ is decreasing and $g'''(x) > 0$, $g'(x)$ is concave up.” This does not earn the first point, because it implies that “ $g'(x)$ is decreasing” is part of the condition on the concavity of $g'(x)$.

· “ $g'(x)$ is decreasing. Since $g'''(x) > 0$, then $g'(x)$ is concave up.” This does earn the first point, because “ $g'(x)$ is decreasing” is not part of the necessary conditions.

· “Since $g'''(x) > 0$, $g'(x)$ is concave up. Since it is also decreasing, the approximation is an underestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is decreasing” is part of the rationale for the answer point.

· “The approximation is an underestimate because $g'(x)$ is decreasing and concave up.” This earns only the first point, because it implies that “ $g'(x)$ is decreasing” is part of the rationale for the answer point.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Concavity
- ☐ Answer

Solution:

Because $g'''(x) > 0$ for $4 \leq x \leq 4.8$, g' is concave up on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_4^{4.8} g'(x) dx$ is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $5 + 70 \cdot 0.4 = 33$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 45 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $5 + 70 \cdot 0.4 + 30 \cdot 0.4$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
4	5	70	0.4	28	33
4.4	33	30	0.4	12	45

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(4.4) \approx g(4) + g'(4)(4.4 - 4) = 5 + 70 \cdot 0.4 = 33$$

$$g(4.8) \approx g(4.4) + g'(4.4)(4.8 - 4.4) \approx 33 + 30 \cdot 0.4 = 45$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 4$.

Because $g''(4) = -120$ and $g'''(4) = 100$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(4)$ and $g'''(4)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(4) = 5$ given in the initial statement of the problem and $g'(4) = 70$ from the table for the coefficients involving those values. For example:

· The expression $g(4) + g'(4)(x - 4) + \frac{g''(4)}{2!}(x - 4)^2 + \frac{g'''(4)}{3!}(x - 4)^3$ earns only one point.

· The expression $5 + 70(x - 4) + \frac{g''(4)}{2!}(x - 4)^2 + \frac{g'''(4)}{3!}(x - 4)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names ($t, x, y, \dots$) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “+ ...” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 4$ will earn no points.



0	1	2
---	---	---

3-1-1a

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(4) + g'(4)(x-4) + \frac{g''(4)}{2!}(x-4)^2 + \frac{g'''(4)}{3!}(x-4)^3 \\ = 5 + 70(x-4) + \frac{-120}{2}(x-4)^2 + \frac{100}{6}(x-4)^3 \\ = 5 + 70(x-4) - 60(x-4)^2 + \frac{50}{3}(x-4)^3 \end{aligned}$$

3-1-1a

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.1	3.2	3.3	3.4
$g'(x)$	14	25	34	41	46

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 10$. Selected values of g' are given in the table above. The function g is three-times differentiable.

- Use the local linear approximation of g at $x = 3$ to approximate $g(3.4)$.
- Find the average rate of change of g' over the interval $3 \leq x \leq 3.4$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = (g(x))^3 - 1$. Find $h'(3)$. Show the computations that lead to your answer.
- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_3^{3.4} g'(x) dx$. Show the computations that lead to your answer.
- It is known that $g'''(x) < 0$ for $3 \leq x \leq 3.4$. Is the approximation from part (d) an overestimate or an underestimate for $\int_3^{3.4} g'(x) dx$? Give a reason for your answer.
- Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.4)$. Show the computations that lead to your answer.
- It is known that $g''(3) = 120$ and $g'''(3) = -200$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

10 + 14 · 0.4. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is 10 + 14 · 0.4.

This numerical response does not need to be simplified but if simplification is attempted, it must be correct to earn this point.

3-1-1a



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3.4) \approx g(3) + g'(3)(3.4 - 3) = 10 + 14 \cdot 0.4 = 15.6$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We must see at a minimum a difference in either the numerator OR denominator before any simplification. For example, $\frac{46-14}{0.4}$ or $\frac{32}{3.4-3}$.

These expressions do not need to be simplified but any additional simplification must be correct.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$\text{Average rate of change} = \frac{g'(3.4) - g'(3)}{3.4 - 3} = \frac{46 - 14}{0.4} = 80$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimal response to earn both points is $3 \cdot 10^2 \cdot 14$.

The first point is for the correct application of the chain rule on the term $(g(x))^3$ to produce the derivative $3 \cdot (g(x))^2 \cdot g'(x)$. If there are any errors in differentiating the constant term, the student does not earn the second point.

3-1-1a

For the first point it is possible that a student expands the expression as a product and uses the product rule instead. If this is done correctly, the student would earn the first point and be eligible for the second point.

The answer point is earned only by the correct answer $3 \cdot 10^2 \cdot 14$. This answer does not need to be simplified but any simplification must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = 3 \cdot (g(x))^2 \cdot g'(x)$$

$$h'(3) = 3 \cdot (g(3))^2 \cdot g'(3) = 3 \cdot 10^2 \cdot 14 = 4200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal response to earn the point is $25 \cdot 0.2 + 41 \cdot 0.2$ or $(25 + 41) \cdot 0.2$.

As always, the numerical answer does not need to be simplified, but if simplification is attempted, it must be correct.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} \int_3^{3.4} g'(x) \, dx &\approx g'(3.1)(3.2 - 3) + g'(3.3)(3.4 - 3.2) \\ &= 25 \cdot 0.2 + 41 \cdot 0.2 = 13.2 \end{aligned}$$

Part E

3-1-1a

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

General:

The first point is for concluding that $g'(x)$ is concave down given that $g'''(x)$ is negative.

The answer point is earned for using that $g'(x)$ is concave down on the given interval to conclude that the midpoint Riemann sum approximation of $\int_3^{3.4} g'(x) dx$ computed in part (d) must necessarily be an overestimate.

Responses that fail to mention $g'(x)$ but instead use ambiguous language such as “the function,” “the graph,” or “it” earn one out of the two points if the response is correct when the ambiguous term is replaced by $g'(x)$.

Responses that only discuss the increasing behavior of $g''(x)$ do not earn either point. The student must bring the discussion down to a consideration about $g'(x)$.

Responses that conclude the approximation is an underestimate cannot earn the answer point.

First point:

We are looking for the response to reach a conclusion regarding $g'(x)$ based on the information they are given regarding $g'''(x)$.

Responses must involve the concavity of $g'(x)$ either by explicitly using the word “concavity” in connection with $g'(x)$ or implicitly by clearly discussing the concavity of $g'(x)$ in graphical terms. For example:

- “Since $g'''(x) < 0$, then $g''(x)$ is decreasing so...” does not earn the first point, because it does not discuss the concavity of $g'(x)$.
- “Since $g'''(x) < 0$, the tangent line to the graph of $g'(x)$ lies above $g'(x)$ so...” does earn the first point, because it discusses the concavity of $g'(x)$ implicitly.

Responses often conclude from the table of values that $g'(x)$ is increasing, and we are willing to overlook this assertion (which may or may not be valid) unless their discussion implies that this is part of the condition of the conclusion. This can impact whether a response earns the first and/or second points. For example:

- “Since $g'(x)$ is increasing and $g'''(x) < 0$, $g'(x)$ is concave down.” This does not earn the first point, because it implies that “ $g'(x)$ is increasing” is part of the condition on the concavity of $g'(x)$.
- “ $g'(x)$ is increasing. Since $g'''(x) < 0$, then $g'(x)$ is concave down.” This does earn the first point, because “ $g'(x)$ is increasing” is not part of the necessary conditions.
- “Since $g'''(x) < 0$, $g'(x)$ is concave down. Since it is also increasing, the approximation is an overestimate.” This earns only the first point, because “it” is inferred to mean $g'(x)$ and this implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.

3-1-1a

· “The approximation is an overestimate because $g'(x)$ is increasing and concave down.” This earns only the first point, because it implies that “ $g'(x)$ is increasing” is part of the rationale for the answer point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Concavity

☐ Answer

Solution:

Because $g'''(x) < 0$ for $3 \leq x \leq 3.4$, g' is concave down on this interval.

Therefore, the midpoint Riemann sum approximation for $\int_3^{3.4} g'(x) dx$ is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler’s method, presented either in a table or in the form shown in the model solution. This requires that there is at most one incorrect value in the first step of the approximation $10 + 14 \cdot 0.2 = 12.8$.

Once a response earns the first point, any subsequent errors will mean the answer point is not earned.

The answer point is earned only if the first point was earned, and the numerical answer of 19.6 or an equivalent is reported.

The numerical answer does not need to be simplified. For example, $10 + 14 \cdot 0.2 + 34 \cdot 0.2$ would earn both points.

Tabular responses:

Tabular responses are common and will generally be some variant of the following (although not all of the columns need to necessarily be included):

x	y	$g'(x)$	Δx	Δy	$y + \Delta y$
3	10	14	0.2	2.8	12.8
3.2	12.8	34	0.2	6.8	19.6

In the presence of correct values in the table above, the column headings do not need to be labeled. However, any error in an unlabeled table will earn no points in this problem.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.2) \approx g(3) + g'(3)(3.2 - 3) = 10 + 14 \cdot 0.2 = 12.8$$

$$g(3.4) \approx g(3.2) + g'(3.2)(3.4 - 3.2) \approx 12.8 + 34 \cdot 0.2 = 19.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for producing any two terms of the third-degree Taylor polynomial, and the second point is for combining these with the remaining terms into a third-degree polynomial centered at $x = 3$.

Because $g''(3) = 120$ and $g'''(3) = -200$ are given explicitly in the statement of the question in part (g), the student need not substitute these numerical values for $g''(3)$ and $g'''(3)$ in the coefficients of the Taylor polynomial expression. However, the student must use the numerical values $g(3) = 10$ given in the initial statement of the problem and $g'(3) = 14$ from the table for the coefficients involving those values. For example:

· The expression $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns only one point.

· The expression $10 + 14(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$ earns both points.

Coefficients that are presented in decimal form must be correct to three decimal places, rounded or truncated.

All variable names ($t, x, y, \dots$) are acceptable as long as that variable is used correctly in all non-constant terms.

Coefficients need not be simplified.

Additional terms or “ $+\dots$ ” fail to earn the second point.

Taylor polynomials that are centered at a point other than $x = 3$ will earn no points.



0	1	2
---	---	---

3-1-1a

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} & g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ &= 10 + 14(x-3) + \frac{120}{2}(x-3)^2 + \frac{-200}{6}(x-3)^3 \\ &= 10 + 14(x-3) + 60(x-3)^2 - \frac{100}{3}(x-3)^3 \end{aligned}$$

3-1-1a

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x .

- (a) Let h be the function defined by $h(x) = f(g(x))$. Find $h'(7)$. Show the work that leads to your answer.
- (b) Let k be a differentiable function such that $k'(x) = (f(x))^2 \cdot g(x)$. Is the graph of k concave up or concave down at the point where $x = 4$? Give a reason for your answer.
- (c) Let m be the function defined by $m(x) = 5x^3 + \int_0^x f'(t) \, dt$. Find $m(2)$. Show the work that leads to your answer.
- (d) Is the function m defined in part (c) increasing, decreasing, or neither at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = f'(g(x)) \cdot g'(x)$ or $h'(7) = f'(g(7)) \cdot g'(7)$.

If the first point is earned, the second point is earned only for an answer of 12 (or equivalent).

If the first point is not earned, the second point can be earned only for a response of either $f'(0) \cdot 8 = 12$ or $\frac{3}{2} \cdot 8$.

A response of 12 with no supporting work does not earn either point.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = f'(g(x)) \cdot g'(x)$$

$$h'(7) = f'(g(7)) \cdot g'(7) = f'(0) \cdot 8 = \frac{3}{2} \cdot 8 = 12$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$.

The first point is also earned by any of the following incorrect expressions, each of which has a single error in the application of the product rule or the chain rule:

- $2f(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f(x) \cdot f'(x) \cdot g'(x)$ or $2f(4) \cdot f'(4) \cdot g'(4)$
- Note: A response that presents one of these expressions cannot earn the second point.

To earn the second point a response must correctly find $k''(4) = -40$ (or equivalent) with supporting work.

The third point is earned for an answer and reason that are consistent with any declared nonzero value of $k''(4)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

3-1-1a

- ☐ Product or chain rule
- ☐ $k''(4)$
- ☐ Answer with reason

Solution:

$$k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$$

$$\begin{aligned} k''(4) &= 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4) \\ &= 2 \cdot 4 \cdot 3 \cdot (-3) + 4^2 \cdot 2 = -72 + 32 = -40 \end{aligned}$$

The graph of k is concave down at the point where $x = 4$ because $k''(4) < 0$ and k'' is continuous.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned only for an answer of 37 (or equivalent) with supporting work equivalent to $5 \cdot 8 + (f(2) - f(0))$, $40 + (f(2) - f(0))$, $5 \cdot 8 + (7 - 10)$, $40 + (7 - 10)$.

An answer of 37 with no supporting work does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with supporting work

Solution:

$$\begin{aligned} m(2) &= 5 \cdot 8 + \int_0^2 f'(t) dt = 40 + (f(2) - f(0)) \\ &= 40 + (7 - 10) = 37 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for considering $m'(x)$, $m'(2)$, or m' . This consideration may appear in a justification statement.

The second point is earned for $m'(2) = 15 \cdot 2^2 + f'(2)$, $m'(2) = 60 + f'(2)$, or $m'(2) = 60 - 8$ but is not earned for an unsupported response of $m'(2) = 52$.

3-1-1a

The third point is earned for an answer and justification consistent with any declared value of $m'(2)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $m'(x)$
- ☐ $m'(2)$ with supporting work
- ☐ Answer with justification

Solution:

$$m'(x) = 15x^2 + f'(x)$$

$$m'(2) = 15 \cdot 4 + f'(2) = 60 + (-8) = 52$$

The graph of m is increasing at $x = 2$ because $m'(2) > 0$.

3-1-1a

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is defined by $f(x) = \int_0^x (t^2 - 3t - 10) dt$.

- Write an equation for the line tangent to the graph of f at $x = 6$.
- Find all values of x at which f has a critical point. Determine whether f has a relative minimum, a relative maximum, or neither at each of these values. Justify your answers.
- On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.
- Let g be the function defined by $g(x) = f(\tan x)$. Find $g'(\frac{\pi}{4})$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported value of -42 does not earn the first point; the minimal supporting work required is $72 - 54 - 60$.

The second point can be earned without presenting an explicit expression for $f'(x)$ by correctly finding $f'(6)$ with supporting work.

Eligibility for third point: The response has earned the second point and used $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x$ to find $f(6)$. In this case the third point is earned for any form of the equation $y = f(6) + 8(x - 6)$ for the declared value of $f(6)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $f(6)$
- ☐ $f'(x)$

3-1-1a

☐ Equation for tangent line

Solution:

$$f(6) = \int_0^6 (t^2 - 3t - 10) dt = \left(\frac{1}{3}t^3 - \frac{3}{2}t^2 - 10t \right) \Big|_0^6$$

$$= \frac{1}{3} \cdot 6^3 - \frac{3}{2} \cdot 6^2 - 10 \cdot 6 = 72 - 54 - 60 = -42$$

$$f'(x) = x^2 - 3x - 10$$

$$f'(6) = 6^2 - 3 \cdot 6 - 10 = 36 - 18 - 10 = 8$$

An equation for the tangent line is $y = -42 + 8(x - 6)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

If any additional critical points are listed or either critical point is missing, the first point is not earned.

If only one of $x = -2$ or $x = 5$ is listed as a critical point and that point is correctly justified as a relative maximum or relative minimum, the response earns one point.

“The slope changes from positive to negative” is not a sufficient justification for the second point. The response must be clear about what slope is changing.

Justifications that appeal only to $f(x)$ changing from increasing to decreasing (or vice versa) are not sufficient for the second point.

A response that imports from part (a) an incorrect $f'(x)$ of the form $ax^2 + bx + c$ (typically, $f'(x) = x^2 - 3x$) may earn the first point for finding the critical points associated with their $f'(x)$. Note: The incorrect $f'(x)$ must be imported from part (a) to be eligible for the first point. Such a response is not eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Critical points
- ☐ Answers with justification

Solution:

3-1-1a

$$f'(x) = x^2 - 3x - 10 = (x - 5)(x + 2) = 0 \quad \text{and } x = 5 \\ \Rightarrow x = -2$$

f has a relative maximum at $x = -2$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = 5$ because $f'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $2x - 3 < 0$ (or $2x - 3 = 0$ or $2x - 3 > 0$) earns the first point.

To earn the second point, a response must explain that the graph of f is concave down on $(-\infty, \frac{3}{2})$ (or $x < \frac{3}{2}$), because $f''(x) < 0$ on that interval.

The interval may include the right endpoint $x = \frac{3}{2}$.

A response that presents any interval other than $(-\infty, \frac{3}{2})$ does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Find $f''(x)$ and compare to zero
- ☐ Answer with reason

Solution:

$$f''(x) = 2x - 3 < 0 \Rightarrow x < \frac{3}{2}$$

The graph of f is concave down on the interval $(-\infty, \frac{3}{2})$ because $f''(x) < 0$ there.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f'(\tan x) \cdot (\sec^2 x)$.

The answer does not need to be simplified: $(\tan^2(\frac{\pi}{4}) - 3 \tan(\frac{\pi}{4}) - 10)(\sec^2(\frac{\pi}{4}))$ or equivalent earns the second point.

3-1-1a



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$$g'(x) = f'(\tan x) \cdot (\sec^2 x) = (\tan^2 x - 3 \tan x - 10)(\sec^2 x)$$

$$\begin{aligned} g'\left(\frac{\pi}{4}\right) &= \left(\tan^2\left(\frac{\pi}{4}\right) - 3 \tan\left(\frac{\pi}{4}\right) - 10\right)(\sec^2\left(\frac{\pi}{4}\right)) \\ &= (1 - 3 - 10)(2) = -24 \end{aligned}$$

3-1-1a

18. Let f be the function defined by $f(x) = \sin^2 x - \sin x$ for $0 \leq x \leq 3\pi/2$. Find the intervals on which f is increasing.

Part B

1 point is earned for $f'(x)$

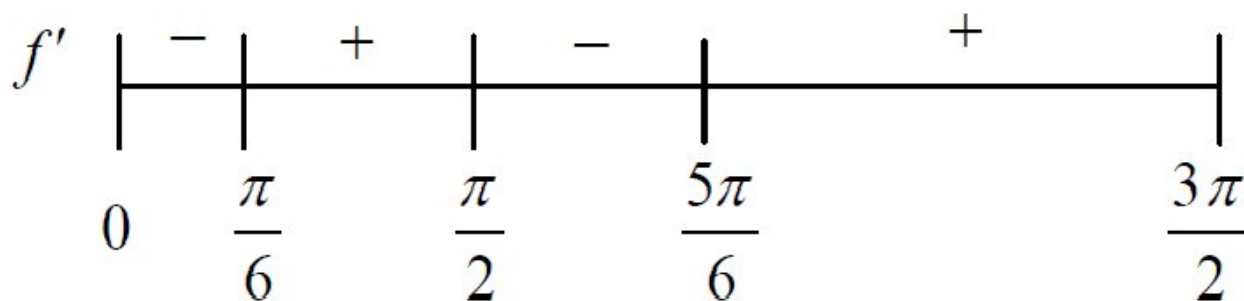
1 point is earned for finding 2 interior critical points

1 point is earned for analysis

1 point is earned for answer

$$\begin{aligned} f'(x) &= 2 \sin x \cos x - \cos x \\ &= \cos x (2 \sin x - 1) \end{aligned}$$

$$f'(x) = 0 \text{ when } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $f'(x)$

1 point is earned for finding 2 interior critical points

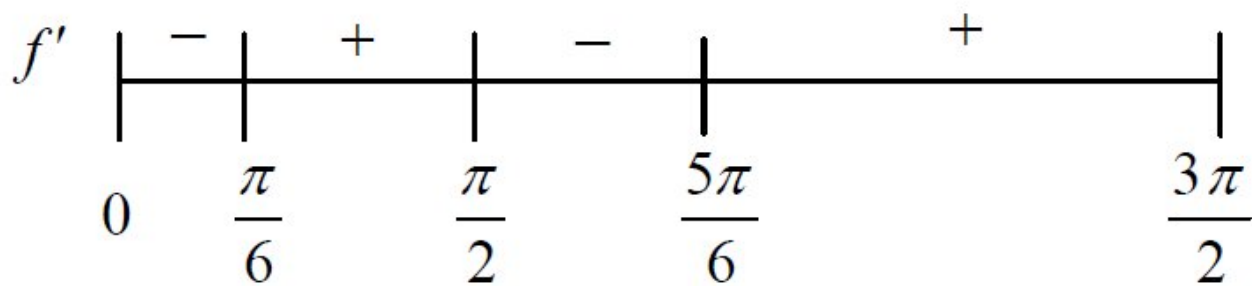
1 point is earned for analysis

3-1-1a

1 point is earned for answer

$$\begin{aligned}f'(x) &= 2 \sin x \cos x - \cos x \\&= \cos x (2 \sin x - 1)\end{aligned}$$

$$f'(x) = 0 \text{ when } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$

3-1-1a

19. Let f be the function defined by $f(x) = \sin^2 x - \sin x$ for $0 \leq x \leq 3\pi/2$.
- Find the x -intercepts of the graph of f .
 - Find the intervals on which f is increasing.
 - Find the absolute maximum value and the absolute minimum value of f . Justify your answer.

Part A

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$$\sin^2 x - \sin x = 0$$

$$\therefore \sin x = 0 \text{ or } \sin x = 1$$

$$x = 0, \frac{\pi}{2}, \pi$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$$\sin^2 x - \sin x = 0$$

$$\therefore \sin x = 0 \text{ or } \sin x = 1$$

$$x = 0, \frac{\pi}{2}, \pi$$

Part B

1 point is earned for $f'(x)$

3-1-1a

1 point is earned for finding 2 interior critical points

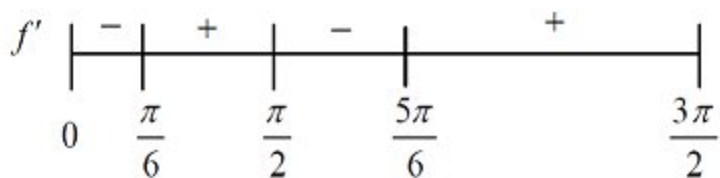
1 point is earned for analysis

1 point is earned for answer

$$f'(x) = 2 \sin x \cos x - \cos x$$

$$= \cos x(2 \sin x - 1)$$

$$f'(x) = 0 \text{ where } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for finding 2 interior critical points

1 point is earned for analysis

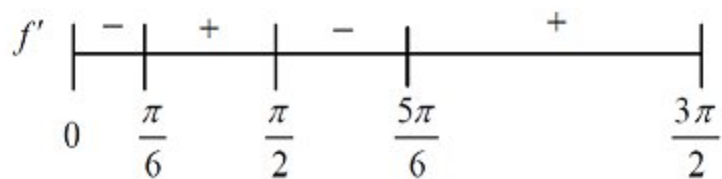
1 point is earned for answer

$$f'(x) = 2 \sin x \cos x - \cos x$$

$$= \cos x(2 \sin x - 1)$$

$$f'(x) = 0 \text{ where } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

3-1-1a



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$

Part C

1 point is earned for max value

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value : $-1/4$

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for max value

3-1-1a

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value : $-1/4$

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2

3-1-1a

Let f be a twice-differentiable function such that $f(2)=5$ and $f(5)=2$. Let g be the function given by $g(x)=f(f(x))$.

20. Show that if $f''(x)=0$ for all x , then the graph of g does not have a point of inflection.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x)=0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) * f'(x) * f'(x) + f'(f(x)) * f''(x)$$

If $f''(x)=0$ for all x , then

$$g''(x) = 0 * f'(x) * f'(x) + f'(f(x)) * 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points.

OR

If $f''(x)=0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned if considers g''

1 point is earned for $g''(x)=0$ for all x

OR

1 point is earned for f is linear

1 point is earned for g is linear

$$g''(x) = f''(f(x)) * f'(x) * f'(x) + f'(f(x)) * f''(x)$$

If $f''(x)=0$ for all x , then

3-1-1a

$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0$ for all x .

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points.

OR

If $f'(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

3-1-1a

21. Show that $g'(2)=g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k)=0$.

Part B

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2)=g'(5)$. Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem

applied to g' on $[2, 5]$ guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2)=g'(5)$.

3-1-1a

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$

guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.

3-1-1a

22. Let f be a twice-differentiable function such that $f(2) = 5$ and $f(5) = 2$. Let g be the function given by $g(x) = f(f(x))$.
- (a) Explain why there must be a value c for $2 < c < 5$ such that $f'(c) = -1$.
- (b) Show that $g'(2) = g'(5)$. Use this result to explain why there must be a value k for $2 < k < 5$ such that $g''(k) = 0$.
- (c) Show that if $f''(x) = 0$ for all x , then the graph of g does not have a point of inflection.
- (d) Let $h(x) = f(x) - x$. Explain why there must be a value r for $2 < r < 5$ such that $h(r) = 0$.

Part A

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{f(5) - f(2)}{5 - 2}$

1 point is earned for conclusion, using MVT

The Mean Value Theorem guarantees that there is a value c , with $2 < c < 5$, so that

$$f'(c) = \frac{f(5) - f(2)}{5 - 2} = \frac{2 - 5}{5 - 2} = -1.$$

Part B

1 point is earned for $g'(x)$

3-1-1a

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$

guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g'(2) = f'(5) \cdot f'(2) = g'(5)$

1 point is earned if uses MVT with g'

$$g'(x) = f'(f(x)) \cdot f'(x)$$

$$g'(2) = f'(f(2)) \cdot f'(2) = f'(5) \cdot f'(2)$$

$$g'(5) = f'(f(5)) \cdot f'(5) = f'(2) \cdot f'(5)$$

Thus, $g'(2) = g'(5)$.

Since f is twice-differentiable, g' is differentiable everywhere, so the Mean Value Theorem applied to g' on $[2, 5]$

guarantees there is a value k , with $2 < k < 5$, such that $g''(k) = \frac{g'(5) - g'(2)}{5 - 2} = 0$.

Part C

1 point is earned if considers g''

1 point is earned for $g''(x) = 0$ for all x

3-1-1a

OR

1 point is earned for f is linear1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. ORIf $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers g'' 1 point is earned for $g''(x) = 0$ for all x

OR

1 point is earned for f is linear1 point is earned for g is linear

$$g''(x) = f''(f(x)) \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot f''(x)$$

If $f''(x) = 0$ for all x , then

$$g''(x) = 0 \cdot f'(x) \cdot f'(x) + f'(f(x)) \cdot 0 = 0 \text{ for all } x.$$

Thus, there is no x -value at which $g''(x)$ changes sign, so the graph of g has no inflection points. ORIf $f''(x) = 0$ for all x , then f is linear, so $g = f \circ f$ is linear and the graph of g has no inflection points.**Part D**1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

3-1-1a

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(2)$ and $h(5)$

1 point is earned for conclusion, using IVT

Let $h(x) = f(x) - x$.

$$h(2) = f(2) - 2 = 5 - 2 = 3$$

$$h(5) = f(5) - 5 = 2 - 5 = -3$$

Since $h(2) > 0 > h(5)$, the Intermediate Value Theorem guarantees that there is a value r , with $2 < r < 5$, such that $h(r) = 0$.

3-1-1a

23. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .

Find the x -coordinate of each critical point of F . For each such x , determine whether $f(x)$ is a relative maximum, a relative minimum, or neither.

Part B

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x=1$

relative maximum at $x=1$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x=1$

relative maximum at $x=1$

3-1-1a

24. Let f be the function given by $f(x) = \ln|x/1+x^2|$

At what values of x does f have a relative maximum or a relative minimum? For each such x , use the first derivative test to determine whether $f(x)$ is a relative maximum or a relative minimum.

Part C

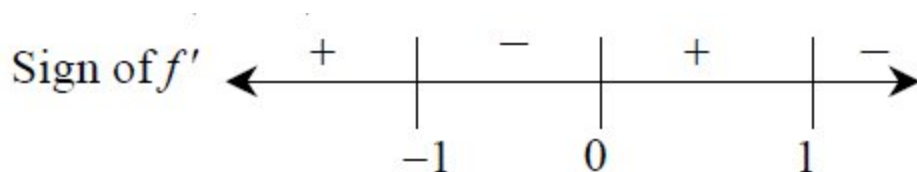
2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$f'(-x) = \frac{1-x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

2 points are earned for the derivative

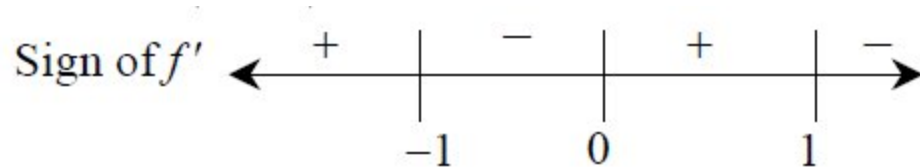
1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$f'(-x) = \frac{1-x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$

3-1-1a



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$

3-1-1a

25. Let f be the function given by $f(x) = \ln|x/1 + x^2|$.

(a) Find the domain of f .

(b) Determine whether f is an even function, an odd function, or neither. Justify your conclusion.

(c) At what values of x does f have a relative maximum or a relative minimum? For each such x , use the first derivative test to determine whether $f(x)$ is a relative maximum or a relative minimum.

(d) Find the range of f .

Part A

1 point is earned for answer

$$x \neq 0$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$x \neq 0$$

Part B

1 point is earned for the answer

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln \left| \frac{-x}{1 + (-x)^2} \right| = f(x)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

3-1-1a

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln \left| \frac{-x}{1 + (-x)^2} \right| = f(x)$$

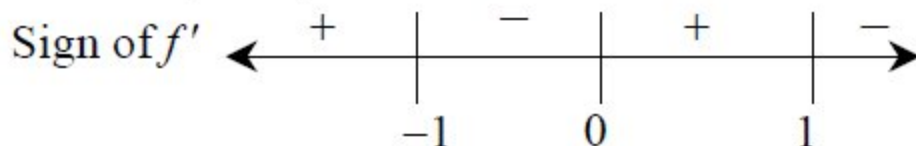
Part C

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$\begin{aligned} f'(x) &= \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2} \\ &= \frac{1-x^2}{x(1+x^2)} \end{aligned}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the derivative

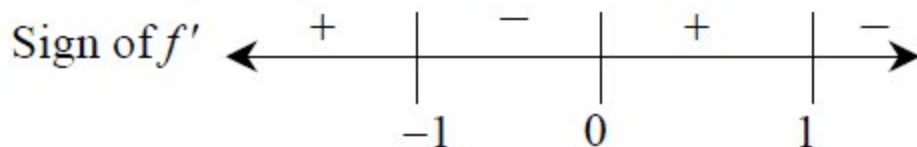
1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

3-1-1a

$$f'(x) = \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$

Part D

2 points are earned for the answer

1 point **deducted** for error

max is

$$f(1) = \ln \frac{1}{2} = -\ln 2$$

$$\lim_{x \rightarrow 0^+} f(x) = -\infty$$

$$\left(\text{or } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = -\infty \right)$$

$$\therefore \text{range} = \left(-\infty, \ln \frac{1}{2} \right]$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the answer

1 point **deducted** for error

max is

3-1-1a

$$f(1) = \ln \frac{1}{2} = -\ln 2$$

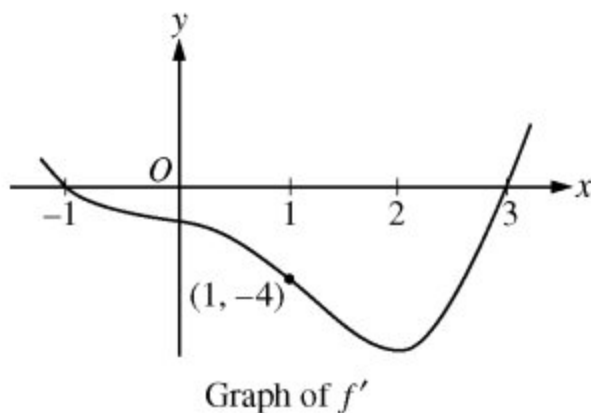
$$\lim_{x \rightarrow 0^+} f(x) = -\infty$$

$$\left(\text{or } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = -\infty \right)$$

$$\therefore \text{range} = \left(-\infty, \ln \frac{1}{2} \right]$$

3-1-1a

26.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

Write an equation for the line tangent to the graph of g at $x = 1$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

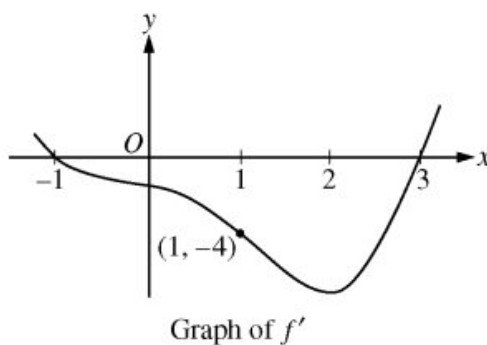
$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.

3-1-1a

3-1-1a

27.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

- Write an equation for the line tangent to the graph of g at $x = 1$.
- For $-1.2 < x < 3.2$, find all values of x at which g has a local maximum. Justify your answer.
- The second derivative of g is $g''(x) = e^{f(x)} [(f'(x))^2 + f''(x)]$. Is $g''(-1)$ positive, negative, or zero? Justify your answer.
- Find the average rate of change of g' , the derivative of g , over the interval $[1, 3]$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

3-1-1a

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.

Part B

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So, g' changes from positive to negative only when f' changes from positive to negative. This occurs at $x = -1$ only. Thus, g has a local maximum at $x = -1$.

Part C

3-1-1a

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[(f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since f' is decreasing on a neighborhood of -1 , $f''(-1) < 0$. Therefore, $g''(-1) < 0$.

Part D

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)} f'(3) - e^{f(1)} f'(1)}{2} = 2e^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

3-1-1a

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)} f'(3) - e^{f(1)} f'(1)}{2} = 2e^2$$

3-1-1a

28. Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x} (3f(x) + 2f'(x))$ for all x . Show that $g''(x) = e^{-2x} (-6f(x) - f'(x) + 2f''(x))$. Does g have a local maximum at $x = 0$? Justify your answer.

Part D

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.

3-1-1a

29.  If $f(x) = \ln(e^{2x})$, then $f'(x) =$

(A) 1

(B) 2



(C) $2x$

(D) e^{-2x}

(E) $2e^{-2x}$

30. If $f'(x) = \frac{(x-2)^3(x^2-4)}{16}$ and $g(x) = f(x^2 - 1)$, what is $g'(2)$?

(A) 2

(B) $\frac{5}{4}$



(C) $\frac{5}{8}$

(D) $\frac{5}{16}$

(E) 0

31. If $f(x) = ae^{-ax}$ for $a > 0$, then $f'(x) =$

(A) e^{-ax}

(B) ae^{-ax}

(C) a^2e^{-ax}

(D) $-ae^{-ax}$

(E) $-a^2e^{-ax}$



3-1-1a

32.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-2	-6	9	-10	16
1	5	-3	3	-2
3	0	7	8	3

The table above gives values of f , f' , g , and g' for selected values of x . If $h(x) = f(g(x))$, what is the value of $h'(1)$?

(A) -19

(B) -14

(C) 7

(D) 9

Answer B

This option is correct. The solution is found by using the chain rule correctly and then using values of the functions and their derivatives from the table to evaluate the derivative at $x = 1$.

$$h'(1) = f'(g(1)) g'(1)$$

$$h'(1) = f'(3) g'(1)$$

$$h'(1) = 7(-2) = -14$$

33.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	6	5	3	-2
1	3	-3	-1	2
3	1	-2	2	3

The table above gives values of f , f' , g , and g' at selected values of x . If $h(x) = f(g(x))$, then $h'(1) =$

(A) 5

(B) 6

(C) 9

(D) 10

(E) 12

3-1-1a

34. If $f(x) = \cos^2(3x - 5)$, then $f'(x) =$

- (A) $6 \cos(3x - 5)$
- (B) $-3\sin^2(3x - 5)$
- (C) $-2 \sin(3x - 5) \cos(3x - 5)$
- (D) $-6 \sin(3x - 5) \cos(3x - 5)$ ✓

35. If $f(x) = (2x^2 + 5)^7$, then $f'(x) =$

- (A) $7(4x)^6$
- (B) $7(2x^2 + 5)^6$
- (C) $14x^2(2x^2 + 5)^6$
- (D) $28x(2x^2 + 5)^6$ ✓

36. If f is a differentiable function and $y = \sin(f(x^2))$ what is $\frac{dy}{dx}$ when $x = 3$?

- (A) $\cos(f(9))$
- (B) $6\cos(f(9))$
- (C) $f'(9) \cos(f(9))$
- (D) $6f'(9) \cos(f(9))$ ✓

37. If $f(x) = x^2 - 4$ and g is a differentiable function of x , what is the derivative of $f(g(x))$?

- (A) $2g(x)$
- (B) $2g'(x)$
- (C) $2xg'(x)$
- (D) $2g(x)g'(x)$ ✓
- (E) $2g(x) - 4$

38.  Let $f(x) = \int_0^{x^2} \sin t \, dt$. At how many points in the closed interval $[0, \sqrt{\pi}]$ does the instantaneous rate of change of f equal the average rate of change of f on that interval?

- (A) Zero
- (B) One
- (C) Two ✓
- (D) Three
- (E) Four

3-1-1a

39. For $x > 0$, $\frac{d}{dx} \int_1^{\sqrt{x}} \frac{1}{1+t^2} dt =$

(A) $\frac{1}{2\sqrt{x}(1+x)}$ ✓

(B) $\frac{1}{2\sqrt{x}(1+\sqrt{x})}$

(C) $\frac{1}{1+x}$

(D) $\frac{\sqrt{x}}{1+x}$

(E) $\frac{1}{1+\sqrt{x}}$

40. For $x > 0$, $\frac{d}{dx} \left(\int_0^{2x} \ln(t^3 + 1) dt \right) =$

(A) $\ln(x^3 + 1)$

(B) $\ln(8x^3 + 1)$

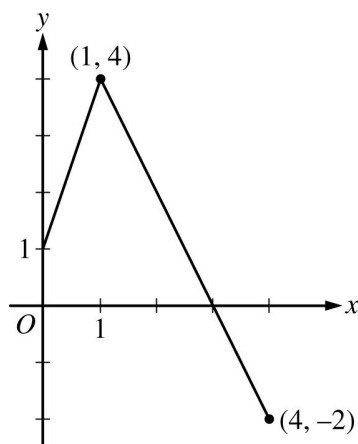
(C) $2 \ln(x^3 + 1)$

(D) $2 \ln(8x^3 + 1)$ ✓

(E) $24x^2 \ln(8x^3 + 1)$

3-1-1a

41.

Graph of f

The graph of the function f , consisting of two line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + 1$, and let h be the function given by $h(x) = f(g(x))$. What is the value of $h'(1)$?

(A) -4



(B) -2

(C) 4

(D) 6

(E) nonexistent

42.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
-1	-5	1	3	0
0	-2	0	1	1
1	0	-3	0	0.5
2	5	-1	5	2

The table above gives values of the differentiable functions f and g and of their derivatives f' and g' , at selected values of x . If $h(x) = f(g(x))$, what is the slope of the graph of h at $x = 2$?

(A) -10

(B) -6

(C) 5

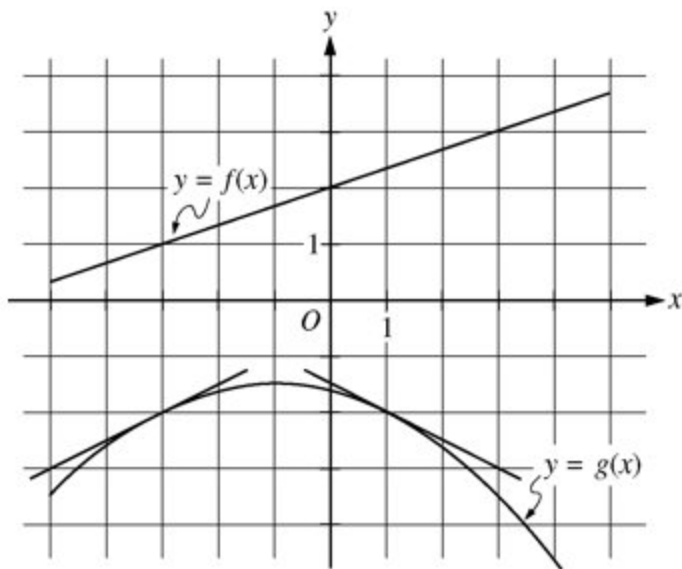
(D) 6



(E) 10

3-1-1a

43.



The figure above shows the graphs of the functions f and g . The graphs of the lines tangent to the graph of g at $x = -3$ and $x = 1$ are also shown. If $B(x) = g(f(x))$, what is $B'(-3)$?

(A) $-\frac{1}{2}$

(B) $-\frac{1}{6}$ ✓

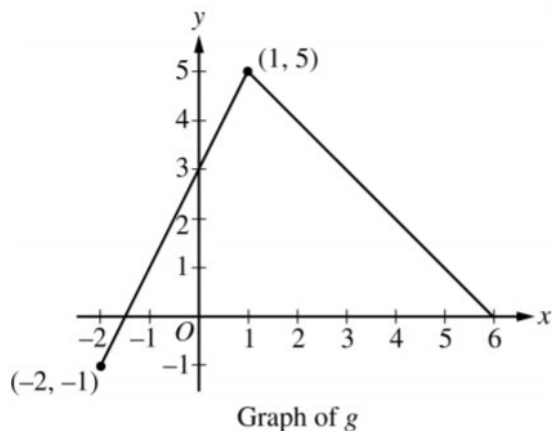
(C) $\frac{1}{6}$

(D) $\frac{1}{3}$

(E) $\frac{1}{2}$

3-1-1a

44.



The graph of the function g is shown above.

If f is the function given by $f(x) = g(g(x))$, what is the value of $f'(0)$?

(A) -2



(B) -1

(C) 2

(D) 3

45. If $y = (x^3 - \cos x)^5$ then $y' =$

(A) $5(x^3 - \cos x)^4$

(B) $5(3x^2 + \sin x)^4$

(C) $5(3x^2 + \sin x)$

(D) $5(3x^2 + \sin x)^4 \cdot (6x + \cos x)$

(E) $5(x^3 - \cos x)^4 \cdot (3x^2 + \sin x)$



3-1-1a

46. The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

(a) Find $f'(x)$.

(b) Write an equation for the line tangent to the graph of f at $x = -3$.

(c) Let g be the function defined by $g(x) = \begin{cases} f(x) & \text{for } -5 \leq x \leq -3 \\ x + 7 & \text{for } -3 < x \leq 5. \end{cases}$

Is g continuous at $x = -3$? Use the definition of continuity to explain your answer.

(d) Find the value of $\int_0^5 x\sqrt{25 - x^2} dx$.

Part A

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$

Part B

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25 - 9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

3-1-1a



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

Part C

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

3-1-1a

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.

Part D

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$

3-1-1a

47. $\frac{d}{dx}(3 + 5x^2)^{1/4} =$

(A) $\frac{1}{4}(3 + 5x^2)^{3/4}$

(B) $\frac{1}{4}(3 + 5x^2)^{3/4}(10x)$

(C) $\frac{1}{4}(3 + 5x^2)^{-3/4}$

(D) $\frac{1}{4}(3 + 5x^2)^{-3/4}(10x)$ ✓

Answer D

Correct. A combination of the power rule and the chain rule is used to compute the derivative.

$$\frac{d}{dx}(3 + 5x^2)^{1/4} = \frac{1}{4}(3 + 5x^2)^{-3/4} \cdot \frac{d}{dx}(3 + 5x^2) = \frac{1}{4}(3 + 5x^2)^{-3/4} \cdot (10x)$$

48. If $y = \cos^2 3x$, then $dy/dx =$

(A) $-6 \sin 3x \cos 3x$ ✓

(B) $-2 \cos 3x$

(C) $2 \cos 3x$

(D) $6 \cos 3x$

(E) $2 \sin 3x \cos 3x$

49.  If $y = \arctan(e^{2x})$, then $\frac{dy}{dx} =$

(A) $\frac{2e^{2x}}{\sqrt{1-e^{4x}}}$

(B) $\frac{2e^{2x}}{1+e^{4x}}$ ✓

(C) $\frac{e^{2x}}{1+e^{4x}}$

(D) $\frac{1}{\sqrt{1-e^{4x}}}$

(E) $\frac{1}{1+e^{4x}}$

3-1-1a

50. If $f(x) = \cos^3(4x)$, then $f'(x) =$

(A) $3\cos^2(4x)$

(B) $-12\cos^2(4x) \sin(4x)$ ✓

(C) $-3\cos^2(4x) \sin(4x)$

(D) $12\cos^2(4x) \sin(4x)$

(E) $-4\sin^3(4x)$

51. If $f(x) = \cos(3x)$, then $f'\left(\frac{\pi}{9}\right) =$

(A) $\frac{3\sqrt{3}}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $-\frac{\sqrt{3}}{2}$

(D) $-\frac{3}{2}$

(E) $-\frac{3\sqrt{3}}{2}$ ✓

52.

x	3	7
$h(x)$	7	22
$h'(x)$	5	10

Selected values of the increasing function h and its derivative h' are shown in the table above.

If g is a differentiable function such that $h(g(x)) = x$ for all x , what is the value of $g'(7)$?

(A) $-1/10$

(B) $1/10$

(C) $1/5$ ✓

(D) $7/5$

3-1-1a

53. Let f be the function given by $f(x) = \sqrt{x^4 - 16x^2}$. Find $f'(x)$.

Part C

The response can earn up to 3 points:

3 points: For the correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) = \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

3 points: For the correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) = \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$

54. If $y = \sin^3 x$, then $\frac{dy}{dx} =$

- (A) $\cos^3 x$
 (B) $3 \cos^2 x$
 (C) $3 \sin^2 x$
 (D) $-3 \sin^2 x \cos x$

(E) $3 \sin^2 x \cos x$



55. If $y = (x/x+1)^5$, then dy/dx

- (A) $5(1+x)^4$
 (B) $\frac{x^4}{(x+1)^4}$
 (C) $\frac{5x^4}{(x+1)^4}$

(D) $\frac{5x^4}{(x+1)^6}$



(E) $\frac{5x^4(2x+1)}{(x+1)^6}$

3-1-1a

56. If $f(x) = e^{1/x}$, then $f'(x) =$

(A) $-\frac{e^{1/x}}{x^2}$ ✓

(B) $-e^{1/x}$

(C) $\frac{e^{1/x}}{x}$

(D) $\frac{e^{1/x}}{x^2}$

(E) $\frac{1}{x}e^{(1/x)-1}$

57. If $f(x) = e^{(2/x)}$, then $f'(x) =$

(A) $2e^{(2/x)} \ln x$

(B) $e^{(2/x)}$

(C) $e^{(-2/x^2)}$

(D) $-\frac{2}{x^2}e^{(2/x)}$ ✓

(E) $-2x^2e^{(2/x)}$

58. If $f(x) = x^2 + 2x$, then $\frac{d}{dx}(f(\ln x)) =$

(A) $\frac{2 \ln x + 2}{x}$ ✓

(B) $2x \ln x + 2x$

(C) $2 \ln x + 2$

(D) $2 \ln x + \frac{2}{x}$

(E) $\frac{2x+2}{x}$

59.  If $f(x) = e^{\tan^2 x}$, then $f'(x) =$

(A) $e^{\tan^2 x}$

(B) $\sec^2 x e^{\tan^2 x}$

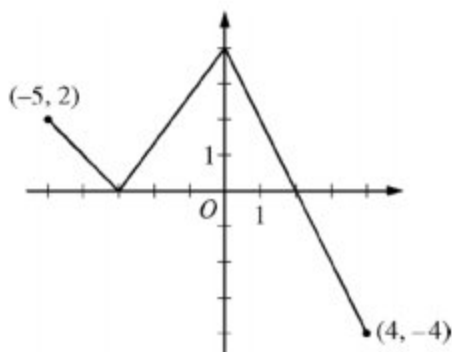
(C) $\tan^2 x e^{\tan^2 x - 1}$

(D) $2 \tan x \sec^2 x e^{\tan^2 x}$ ✓

(E) $2 \tan x e^{\tan^2 x}$

3-1-1a

60.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part D

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f(x^2 - x)(2x - 1) \quad p'(-1) = f(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $p'(x)$

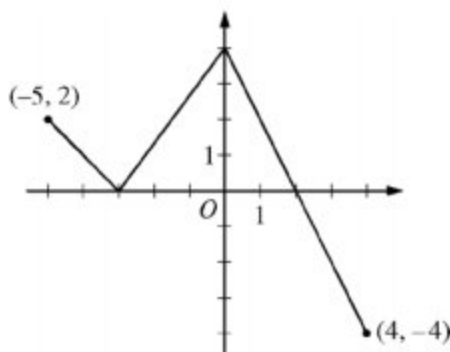
1 point is earned for the answer

$$p'(x) = f(x^2 - x)(2x - 1)$$

$$p'(-1) = f(2)(-3) = (-2)(-3) = 6$$

3-1-1a

61.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

- Find $g(3)$.
- On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.
- The function h is defined by $h(x) = \frac{g(x)}{5x}$. Find $h'(3)$.
- The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part A

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$

3-1-1a

Part B

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$\begin{aligned} h'(3) &= \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2} \\ &= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3-1-1a

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$h'(3) = \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2}$$

$$= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3}$$

Part D

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$

3-1-1a

The twice-differentiable function f is defined for all real numbers and satisfies the following conditions: $f(0) = 2$, $f'(0) = -4$, and $f''(0) = 3$.

62. The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part B

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.

3-1-1a

63. The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a . Show the work that leads to your answer.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = aeax + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a2eax + f''(x)$$

$$g''(0) = a2 + 3$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = aeax + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a2eax + f''(x)$$

$$g''(0) = a2 + 3$$

3-1-1a

64. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions:

$$f(0) = 2, f'(0) = -4, \text{ and } f''(0) = 3.$$

- (a) The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a . Show the work that leads to your answer.
- (b) The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$

Part B

3-1-1a

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

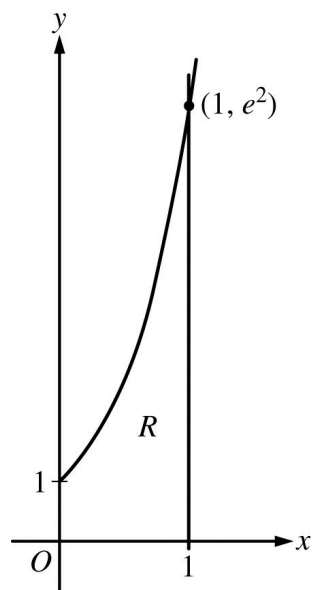
$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0) f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.

3-1-1a

65.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

Write an equation for the line tangent to the graph of f at $x = 1$.

Part A

The question can earn up to 2 points:

1 point: for $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: for the correct answer.

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$



0	1	2
---	---	---

The response earns both of the following points:

1 point: for $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

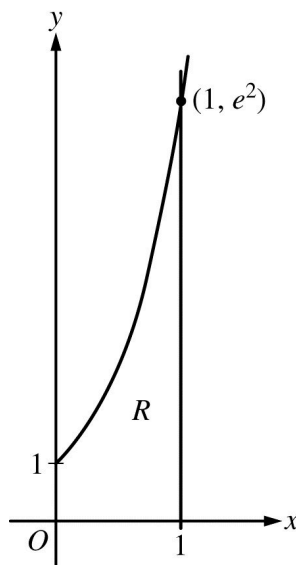
1 point: for the correct answer.

3-1-1a

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$

3-1-1a

66.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

- Write an equation for the line tangent to the graph of f at $x = 1$.
- Find the area of R .
- Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The response can earn up to 2 points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For $f'(1)$

3-1-1a

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.

Part B

The response can earn up to 3 points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$

Part C

The response can earn up to 4 points:

2 points: For the integrand

3-1-1a

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

2 points: For the integrand

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

67. If $f(x) = (x - 1)^{\frac{3}{2}} + \frac{e^{x-2}}{2}$, then $f'(2) =$

(A) 1

(B) $\frac{3}{2}$

(C) 2

(D) $\frac{7}{2}$ (E) $\frac{3+e}{2}$

68. If $f(x) = (x - 1)(x^2 + 2)^3$, then $f'(x) =$

(A) $6x(x^2 + 2)^2$ (B) $6x(x - 1)(x^2 + 2)^2$ (C) $(x^2 + 2)^2(x^2 + 3x - 1)$ (D) $(x^2 + 2)^2(7x^2 - 6x + 2)$ (E) $-3(x - 1)(x^2 + 2)^2$

3-1-1a

69. If $f(x) = \ln(\ln x)$, then $f'(x) =$

- (A) $1/x$
(B) $1/\ln x$
(C) $\ln x/x$
(D) x

(E) $1/x \ln x$ ✓

70. If $y = x^2 - 2x$ and $u = 2x + 1$, then $\frac{dy}{du} =$

- (A) $\frac{2(x^2+x-1)}{(2x+1)^2}$
(B) $6x^2 - 3x - 2$
(C) $4x$

(D) $x - 1$ ✓

(E) $\frac{1}{x-1}$

71. Let f and g be differentiable functions such that

$$f(1) = 2, \quad f'(1) = 3, \quad f'(2) = -4,$$

$$g(1) = 2, \quad g'(1) = -3, \quad g'(2) = 5.$$

If $h(x) = f(g(x))$, then $h'(1) =$

- (A) -9
(B) -4
(C) 0

(D) 12 ✓

(E) 15

72. $\frac{d}{dx} \ln \left| \cos\left(\frac{\pi}{x}\right) \right|$ is

- (A) $\frac{-\pi}{x^2 \cos\left(\frac{\pi}{x}\right)}$
(B) $-\tan\left(\frac{\pi}{x}\right)$
(C) $\frac{1}{\cos\left(\frac{\pi}{x}\right)}$
(D) $\frac{\pi}{x} \tan\left(\frac{\pi}{x}\right)$

(E) $\frac{\pi}{x^2} \tan\left(\frac{\pi}{x}\right)$ ✓

3-1-1a

73. $\frac{d}{dx}(\sin^3(x^2)) =$

- (A) $\cos^3(x^2)$
(B) $3\sin^2(x^2)$
(C) $6x\sin^2(x^2)$
(D) $3\sin^2(x^2)\cos(x^2)$

(E) $6x\sin^2(x^2)\cos(x^2)$ ✓

74. If $f(x) = \ln(x + 4 + e^{-3x})$, then $f'(0)$ is

(A) $-\frac{2}{5}$ ✓

(B) $\frac{1}{5}$

(C) $\frac{1}{4}$

(D) $\frac{2}{5}$

(E) nonexistent

75. If $f(x) = \sin(e^{-x})$, then $f'(x) =$

(A) $-\cos(e^{-x})$

(B) $\cos(e^{-x}) + e^{-x}$

(C) $\cos(e^{-x}) - e^{-x}$

(D) $e^{-x}\cos(e^{-x})$

(E) $-e^{-x}\cos(e^{-x})$ ✓

Answer E

76. If $f(x) = \ln(x + 4 + e^{-3x})$, then $f'(0)$ is

(A) $-\frac{2}{5}$ ✓

(B) $\frac{1}{5}$

(C) $\frac{1}{4}$

(D) $\frac{2}{5}$

(E) nonexistent

3-1-1a

77. If $f(x) = (x^2 - 2x - 1)^{\frac{2}{3}}$, then $f'(0)$ is

(A) $\frac{4}{3}$



(B) 0

(C) $-\frac{2}{3}$

(D) $-\frac{4}{3}$

(E) -2

78. If $f(x) = \sin^2(3 - x)$, then $f'(0) =$

(A) $-2 \cos 3$

(B) $-2 \sin 3 \cos 3$



(C) $6 \cos 3$

(D) $2 \sin 3 \cos 3$

(E) $6 \sin 3 \cos 3$

3-1-1a

79. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 2$. The derivative of f is given by $f'(x) = \sin(\pi x) + x^2 + 3$.

- (a) Find $f''(-2)$.
- (b) Write an equation for the line tangent to the graph of $y = \frac{1}{f(x)}$ at $x = 0$.
- (c) Let g be the function defined by $g(x) = f(\sqrt{3x^2 + 4})$. Find $g'(2)$.
- (d) Let h be the inverse function of f . Find $h'(2)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = \pi \cos(\pi x) + 2x$$

$$f''(-2) = \pi \cos(-2\pi) + 2(-2) = \pi - 4$$

3-1-1a

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error.

The third point may be earned if response contains a slope that is consistent with the derivative expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\frac{d}{dx} \left(\frac{1}{f(x)} \right)$
- ☐ slope
- ☐ equation for tangent line

Solution:

$$\text{At } x = 0, y = \frac{1}{f(0)} = \frac{1}{2}.$$

$$\frac{d}{dx} \left(\frac{1}{f(x)} \right) = \frac{d}{dx} (f(x))^{-1} = -(f(x))^{-2} f'(x) = -\frac{f'(x)}{(f(x))^2}$$

$$\left. \frac{d}{dx} \left(\frac{1}{f(x)} \right) \right|_{x=0} = -\frac{f'(0)}{(f(0))^2} = -\frac{3}{4}$$

An equation for the tangent line is $y = \frac{1}{2} - \frac{3}{4}x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

3-1-1a

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'(x) = f'(\sqrt{3x^2 + 4}) \cdot \frac{6x}{2\sqrt{3x^2 + 4}} = f'(\sqrt{3x^2 + 4}) \cdot \frac{3x}{\sqrt{3x^2 + 4}}$$

$$g'(2) = f'(\sqrt{3 \cdot 2^2 + 4}) \cdot \frac{3 \cdot 2}{\sqrt{3 \cdot 2^2 + 4}} = f'(4) \cdot \frac{3}{2} = 19 \cdot \frac{3}{2} = \frac{57}{2}$$

Part D

Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct value of $h'(2)$

Solution:

Because $f(0) = 2$, $h(2) = 0$.

$$h'(2) = \frac{1}{f'(h(2))} = \frac{1}{f'(0)} = \frac{1}{3}$$

3-1-1a

80. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 3$. The derivative of f is given by $f'(x) = \cos(\pi x) + x^4 + 6$.

- (a) Find $f''(-3)$.
- (b) Write an equation for the line tangent to the graph of $y = (f(x))^2$ at $x = 0$.
- (c) Let g be the function defined by $g(x) = f(\sqrt{2x^2 + 7})$. Find $g'(3)$.
- (d) Let h be the inverse function of f . Find $h'(3)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = -\pi \sin(\pi x) + 4x^3$$

$$f''(-3) = -\pi \sin(-3\pi) + 4(-3)^3 = -108$$

Part B

3-1-1a

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error. Trigonometric function values do not need to be evaluated. The third point may be earned if response contains a slope that is consistent with the derivative expression. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $\frac{d}{dx}((f(x))^2)$
- ☐ slope
- ☐ equation for tangent line

Solution:

At $x = 0$, $y = (f(0))^2 = 3^2 = 9$.

$$\frac{d}{dx}((f(x))^2) = 2f(x)f'(x)$$

$$\left. \frac{d}{dx}((f(x))^2) \right|_{x=0} = 2f(0)f'(0) = 2(3)(\cos 0 + 6) = 6(\cos 0 + 6) = 42$$

An equation for the tangent line is $y = 9 + 42x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. Trigonometric function values do not need to be evaluated. A simplified answer is not required. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

3-1-1a

$$g'(x) = f'(\sqrt{2x^2 + 7}) \cdot \frac{4x}{2\sqrt{2x^2 + 7}} = f'(\sqrt{2x^2 + 7}) \cdot \frac{2x}{\sqrt{2x^2 + 7}}$$

$$g'(3) = f'(\sqrt{2 \cdot 3^2 + 7}) \cdot \frac{2 \cdot 3}{\sqrt{2 \cdot 3^2 + 7}} = f'(5) \cdot \frac{6}{5} = (\cos(5\pi) + 631) \cdot \frac{6}{5} = (630) \frac{6}{5} = 756$$

Part D

Substitution of function values is required. Trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

Because $f(0) = 3$, $h(3) = 0$.

$$h'(3) = \frac{1}{f'(h(3))} = \frac{1}{f'(0)} = \frac{1}{(\cos 0 + 6)} = \frac{1}{7}$$

3-1-1a

81. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	-2	-1	1
$f(x)$	$-\frac{5}{2}$	-3	-2	$\frac{2}{3}$
$f'(x)$	-1	$\frac{1}{3}$	$\frac{6}{5}$	$\frac{4}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{f(x^2)}{e^x}$. Find $g'(-1)$.
- (b) Let h be the function defined by $h(x) = f(f(-2x))$. Find $h'(1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arcsin\left(\frac{x}{2}\right)$. Find $k'(-1)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

3-1-1a**Solution:**

$$g'(x) = \frac{f'(x^2) \cdot 2x \cdot e^x - f(x^2) \cdot e^x}{(e^x)^2}$$

$$\begin{aligned} g'(-1) &= \frac{f'(1) \cdot 2(-1) \cdot e^{-1} - f(1) \cdot e^{-1}}{(e^{-1})^2} \\ &= \frac{\frac{4}{3}(-2)e^{-1} - \frac{2}{3}e^{-1}}{(e^{-1})^2} = -\frac{10e}{3} \end{aligned}$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$\begin{aligned} h'(x) &= f'(f(-2x)) \cdot f'(-2x) \cdot (-2) \\ h'(1) &= f'(f(-2)) \cdot f'(-2) \cdot (-2) \\ &= f'(-3) \cdot \frac{1}{3} \cdot (-2) = (-1) \cdot \frac{1}{3} \cdot (-2) = \frac{2}{3} \end{aligned}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

3-1-1a



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arcsin\left(\frac{x}{2}\right) + f(x) \cdot \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2}$$

$$\begin{aligned} k'(-1) &= f'(-1) \cdot \arcsin\left(\frac{-1}{2}\right) + f(-1) \cdot \frac{1}{\sqrt{1-\left(\frac{-1}{2}\right)^2}} \cdot \frac{1}{2} \\ &= \frac{6}{5} \cdot \left(-\frac{\pi}{6}\right) + (-2) \cdot \frac{2}{\sqrt{3}} \cdot \frac{1}{2} = -\frac{\pi}{5} - \frac{2}{\sqrt{3}} \end{aligned}$$

3-1-1a

82. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	2	3	8
$f(x)$	-9	4	2	6
$f'(x)$	$-\frac{7}{2}$	$\frac{3}{2}$	$-\frac{2}{5}$	$\frac{1}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{\ln x}{f(x^3)}$. Find $g'(2)$.
- (b) Let h be the function defined by $h(x) = f(f(-3x))$. Find $h'(-1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arctan\left(\frac{x}{3}\right)$. Find $k'(-3)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

3-1-1a

$$g'(x) = \frac{f(x^3) \cdot \frac{1}{x} - f'(x^3) \cdot 3x^2 \cdot \ln x}{(f(x^3))^2}$$

$$g'(2) = \frac{f(8) \cdot \frac{1}{2} - f'(8) \cdot 12 \ln 2}{(f(8))^2}$$

$$= \frac{6 \cdot \frac{1}{2} - \frac{1}{3} \cdot 12 \ln 2}{(6)^2} = \frac{3 - 4 \ln 2}{36}$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$h'(x) = f'(f(-3x)) \cdot f'(-3x) \cdot (-3)$$

$$h'(-1) = f'(f(3)) \cdot f'(3) \cdot (-3)$$

$$= f'(2) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \left(\frac{3}{2}\right) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \frac{9}{5}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

3-1-1a



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arctan\left(\frac{x}{3}\right) + f(x) \cdot \frac{1}{1+\left(\frac{x}{3}\right)^2} \cdot \frac{1}{3}$$

$$\begin{aligned} k'(-3) &= f'(-3) \cdot \arctan(-1) + f(-3) \cdot \frac{1}{1+\left(\frac{-3}{3}\right)^2} \cdot \frac{1}{3} \\ &= -\frac{7}{2} \cdot \left(-\frac{\pi}{4}\right) + (-9) \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{7\pi}{8} - \frac{3}{2} \end{aligned}$$

83. If $f(x)=\tan(2x)$, then $f'\left(\frac{\pi}{6}\right) =$

- (A) $\sqrt{3}$
- (B) $2\sqrt{3}$
- (C) 4
- (D) $4\sqrt{3}$

(E) 8



84. If $y = \cos 2x$, then $\frac{dy}{dx} =$

(A) $-2\sin 2x$

(B) $-\sin 2x$

(C) $\sin 2x$

(D) $2 \sin 2x$

(E) $2 \sin x$



3-1-1a

85. If $f(x) = e^{2x}(x^3 + 1)$, then $f'(2) =$

- (A) $6e^4$
- (B) $21e^4$
- (C) $24e^4$
- (D) $30e^4$

**Answer D**

This option is correct. Use a combination of the product rule and the chain rule to compute the derivative, then evaluate that derivative at $x = 2$.

$$f'(x) = 2e^{2x}(x^3 + 1) + e^{2x}(3x^2)$$

$$f'(2) = 18e^4 + 12e^4 = 30e^4$$

86. If $y = x^2 \sin 2x$, then $\frac{dy}{dx} =$

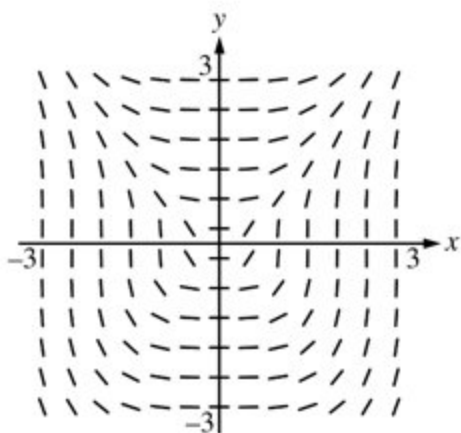
- (A) $2x \cos 2x$
- (B) $4x \cos 2x$
- (C) $2x(\sin 2x + \cos 2x)$
- (D) $2x(\sin 2x - x \cos 2x)$

(E) $2x(\sin 2x + x \cos 2x)$



3-1-1a

87.



Shown above is a slope field for which of the following differential equations?

- (A) $\frac{dy}{dx} = \frac{x}{y}$
- (B) $\frac{dy}{dx} = \frac{x^2}{y^2}$
- (C) $\frac{dy}{dx} = \frac{x^3}{y}$
- (D) $\frac{dy}{dx} = \frac{x^2}{y}$
- (E) $\frac{dy}{dx} = \frac{x^3}{y^2}$ ✓

88. An equation of the line tangent to the graph of $f(x) = x(1 - 2x)^3$ at the point $(1, -1)$ is

- (A) $y = -7x + 6$ ✓
- (B) $y = -6x + 5$
- (C) $y = -2x$
- (D) $y = 2x - 3$
- (E) $y = 7x - 8$

89. $d/dx(\ln e^{2x}) =$

- (A) $1/e^{2x}$
- (B) $2/e^{2x}$
- (C) $2x$
- (D) 1
- (E) 2 ✓